

Mahakumbh BhashaSathi: Multilingual Pilgrim Companion

Project Summary

Mahakumbh BhashaSathi is a lightweight, browser-based assistant designed to help pilgrims at the Mahakumbh festival navigate the event safely and confidently. It combines local guidance, emergency support, accommodation and service details, and multilingual phrase translation into a single accessible interface.

Problem Statement

Mahakumbh attracts visitors from many regions and languages. Pilgrims often face challenges finding event schedules, locating support services, understanding accommodation options, and communicating in a language they know. A simple, offline-capable assistant can reduce confusion and improve safety.

Solution

This project provides a local Python-powered web server and static frontend that runs in a browser without requiring heavy frameworks or external cloud services. Users can ask questions, receive information on navigation, events, services, and emergencies, and translate common pilgrimage phrases across supported languages.

Key Features

- Navigation and route guidance for festival areas
- Event and ritual schedule information
- Emergency contact advice and safety instructions
- Accommodation and local service guidance
- FAQ-style answers for common pilgrim questions
- Phrase translation support across English, Hindi, Bengali, Gujarati, Marathi, Punjabi, Tamil, Telugu, Malayalam, Kannada, French, Japanese, and German
- Offline-friendly implementation using Python standard libraries and static HTML/CSS/JavaScript

Technology Stack

- Python 3.13
- ``http.server`` standard library for backend API
- Static frontend with HTML, CSS, and vanilla JavaScript
- ``reportlab`` used only for generating the submission PDF presentation if needed
- No React, Next.js, Django, Flask, Node.js, or remote database dependencies

Architecture

- ``maha_kumbh_assistant.py``: backend server logic, request handling, response selection, translation engine
- ``static/index.html``: user interface and chat-style interaction
- ``static/style.css``: visual layout and responsive styling
- ``static/app.js``: frontend messaging, form handling, and API integration
- ``tests/test_assistant.py``: unit tests for translation and answer selection logic

Usage Instructions

1. Open a terminal in ``c:\dataset\maha_kumbh_assistant``
2. Activate the virtual environment if available
3. Run:

```
```powershell
python maha_kumbh_assistant.py
```
```
4. Open ``http://127.0.0.1:8080`` in a browser and enter questions or translation prompts.

Testing

Run the provided test suite to verify translation and answer handling:

```
```powershell
python -m unittest discover tests
```

...

### **## Submission Materials**

- `README.md` with setup and usage instructions
- `PROJECT\_SUMMARY.md` for judges and review panels
- `README.pdf` generated from the project documentation
- `tests/test\_assistant.py` for validation purposes

### **## Future Enhancements**

- Expand the multilingual phrase database and translation patterns
- Add dynamic map integration for real-time navigation
- Enhance question answering with a small local model or curated knowledge base
- Support audio input and voice responses for easier pilgrim interaction